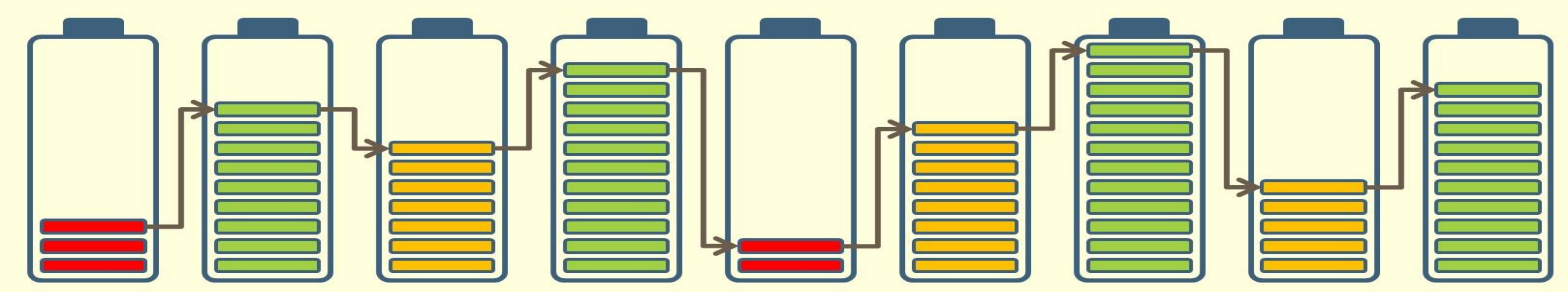
# From materials to vehicle – what, why, and how?
## → From vehicle to materials

Helena Berg



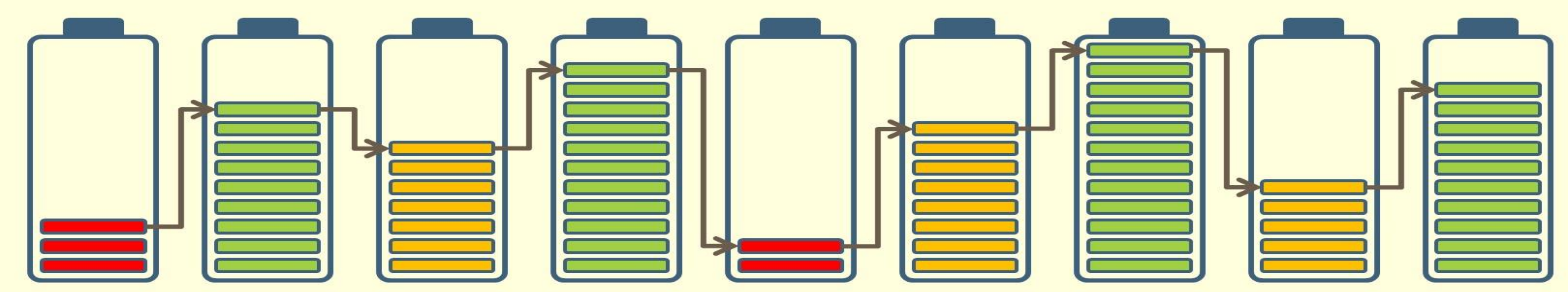
# Outline

1. Electric vehicles and requirements
2. Battery packs for vehicles
3. Cell selection
4. Material requirements
5. Li-ion materials
6. From material to cell
7. From research to production

Libergreen

# 1. Electric vehicles and requirements



# Types of electric vehicles (xEV)

- Start/Stop
- Mild hybrids (HEV)
- Strong (or full) hybrids (HEV)
- Plug-in hybrids (PHEV)
- Full electric (EV, BEV, PEV)
  - PEV can also be FCEV or FCV

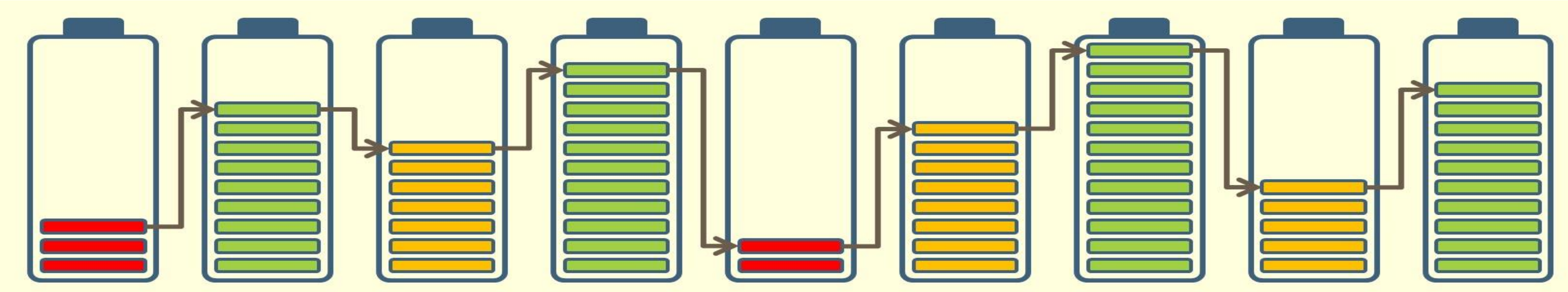
# Vehicle requirements

- Electric driving range
- Vehicle weight
- Roll and drag resistance
- Packaging limitations
- Operational conditions incl. auxillary loads
- Performance (e.g., BMW's 'fun to drive')
- Climate/Geographical constrains
- Durability
- Cost
- Service needs
- Charging

All must be met

# 2. Battery packs for vehicles



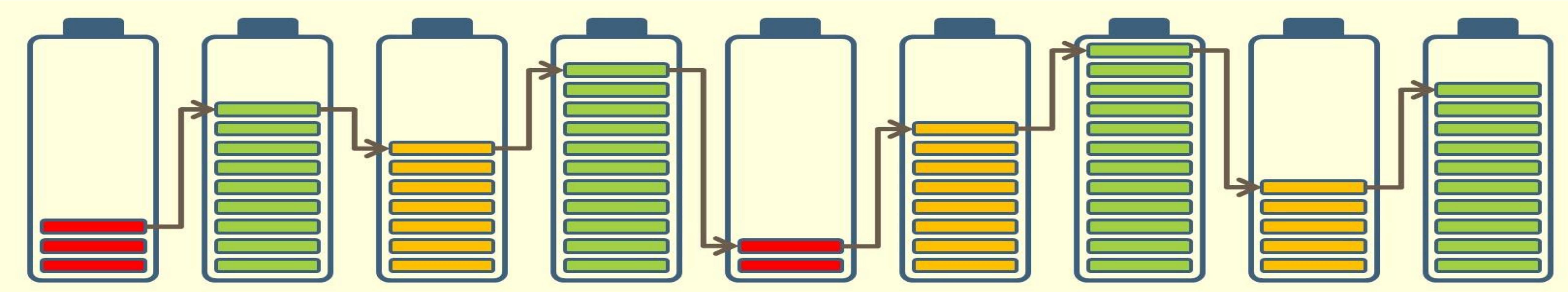
# A battery pack includes….

- Cells
  - Often in modules
  - Connected in Series (and Parallel) –
- Electronics
  - Supervision and Balancing
- Wires for current distribution (often Cu)
- Cooling
  - Liquid or Air; Active or Passive
- Control unit
- Fuses
- Disconnect unit
- Connectors
- Housing and safety protection

Cells 50-75 % of pack weight and ca 75 % of pack cost

# Battery control

- State of Charge (SOC) - minutes
- State of Power (SOP) - seconds
- State of Health (SOH) - months



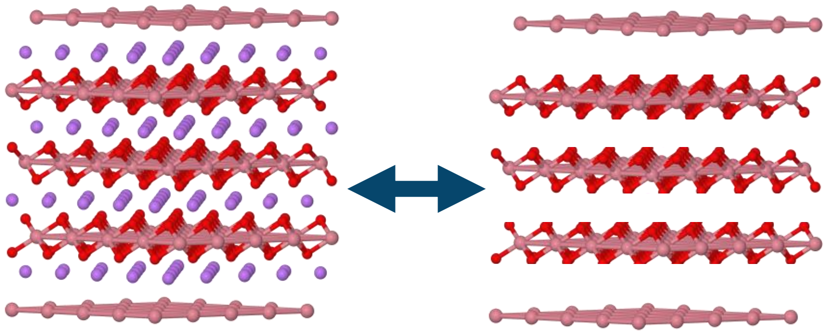




Libergreen

# State of Charge

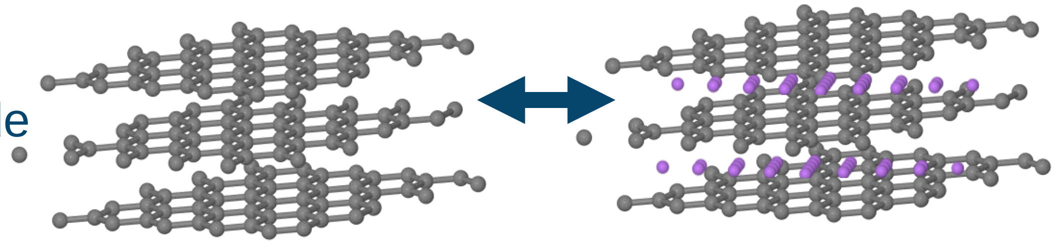
Cathode

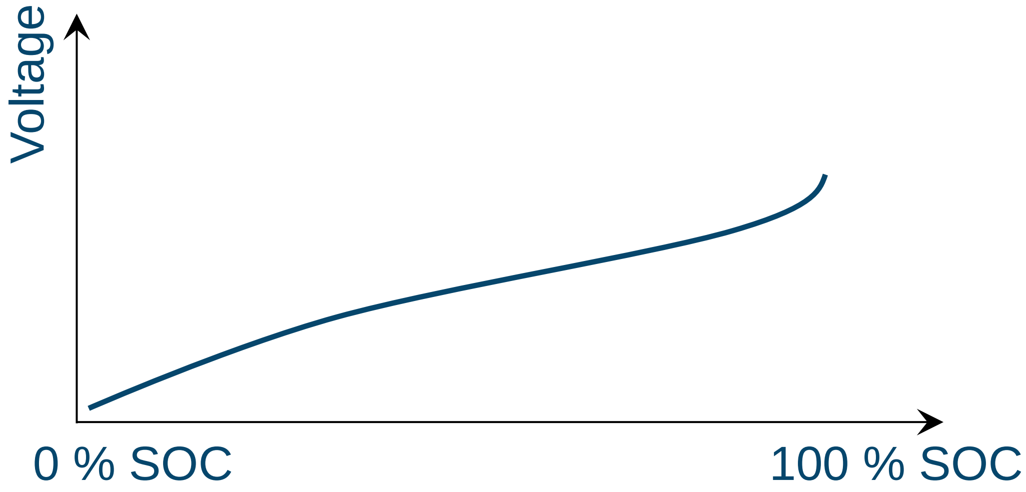


SOC 0 %

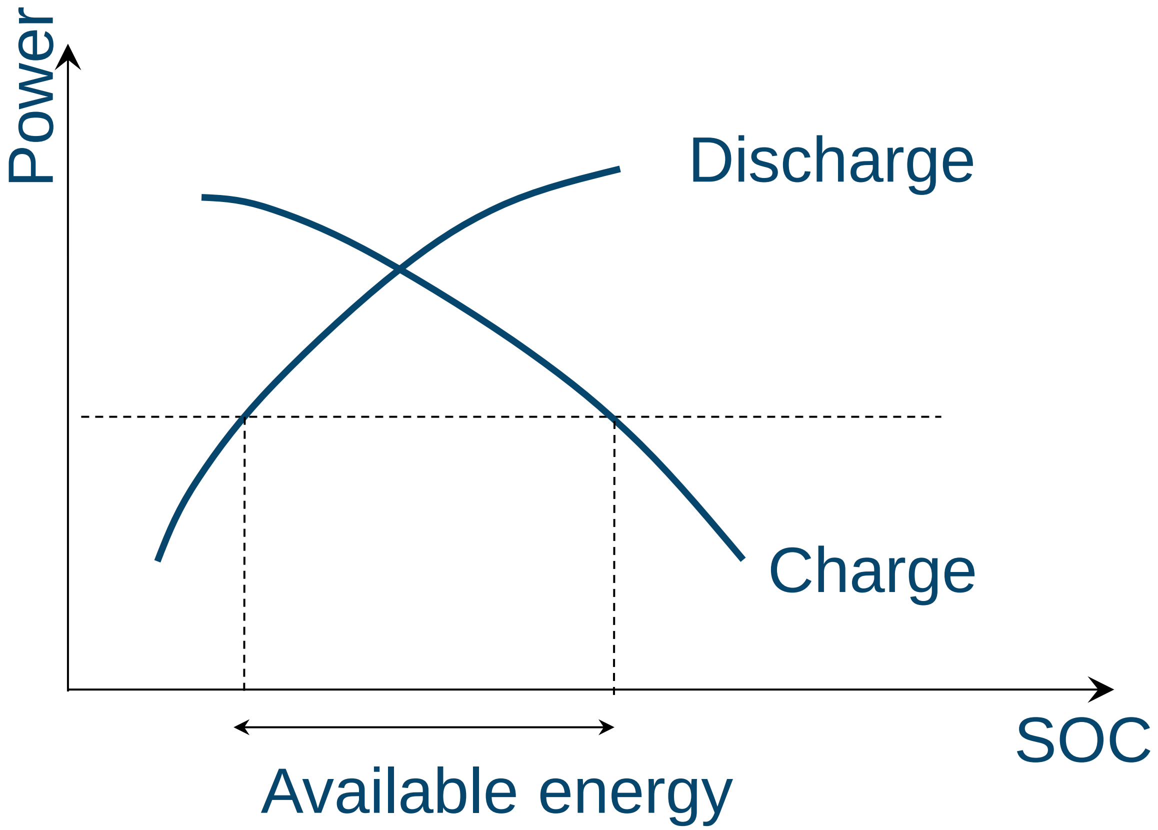
SOC 100 %

Anode





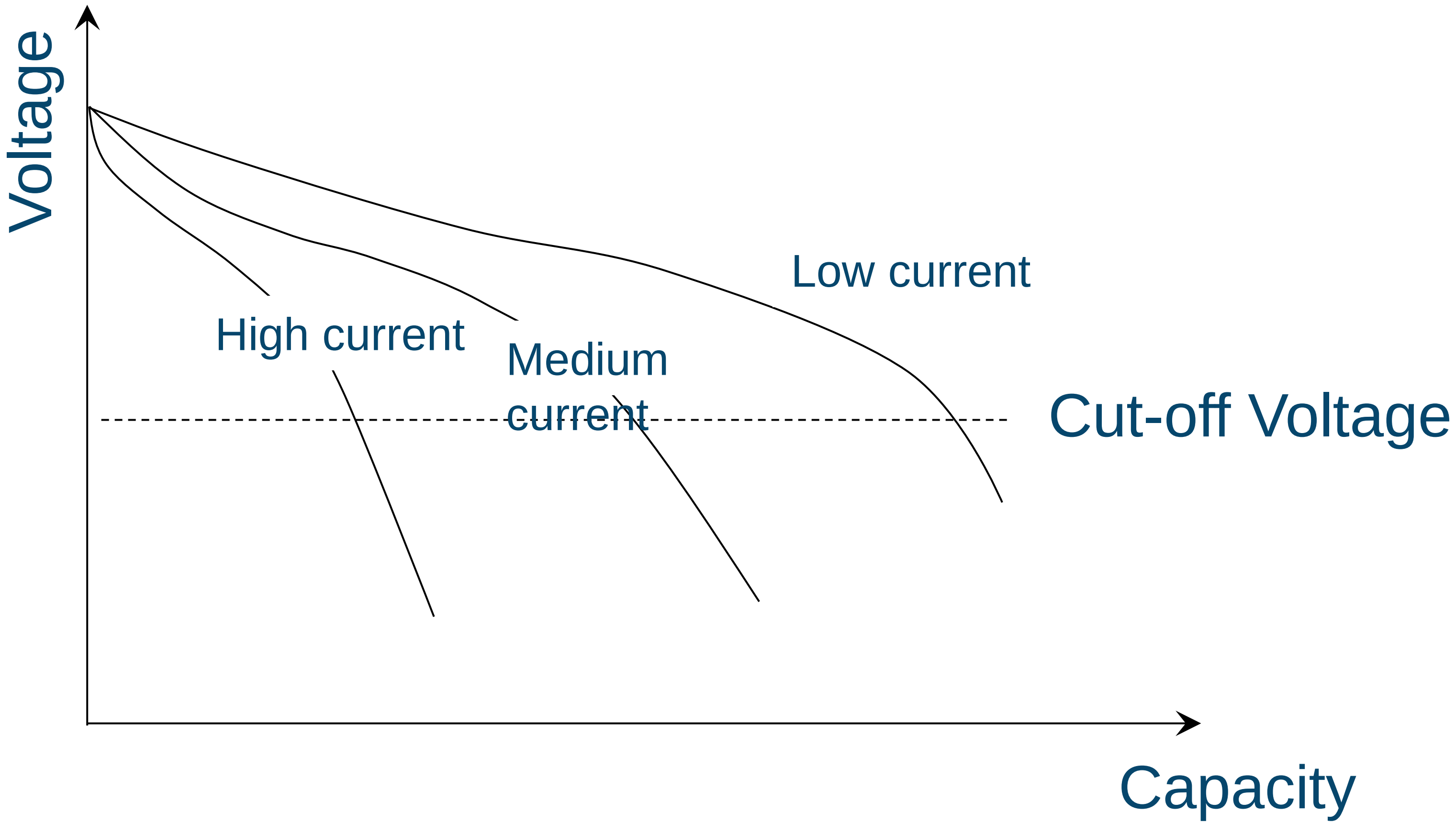
# State of Power vs. State of Charge






Libergreen

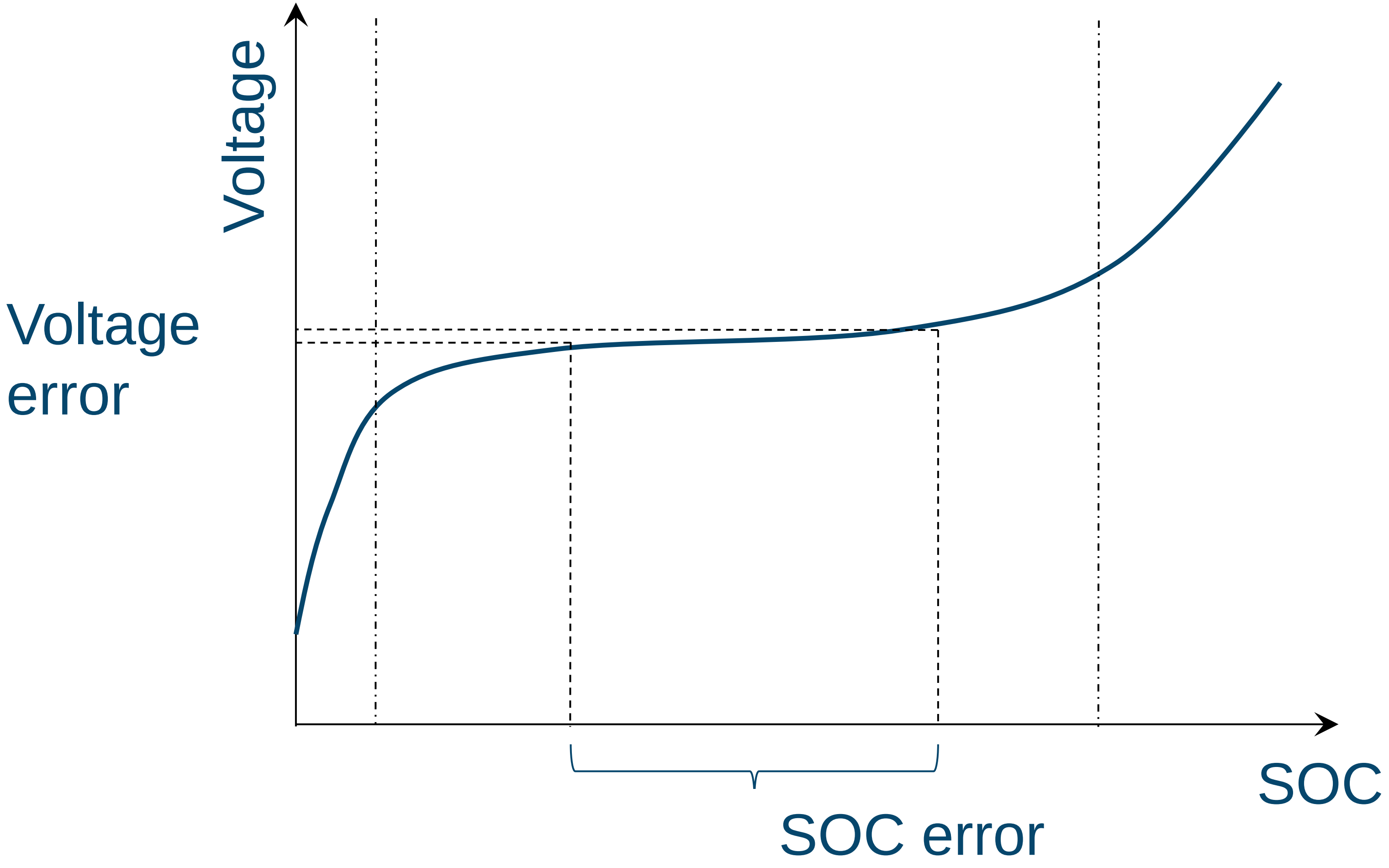
# Current effects



Libergreen

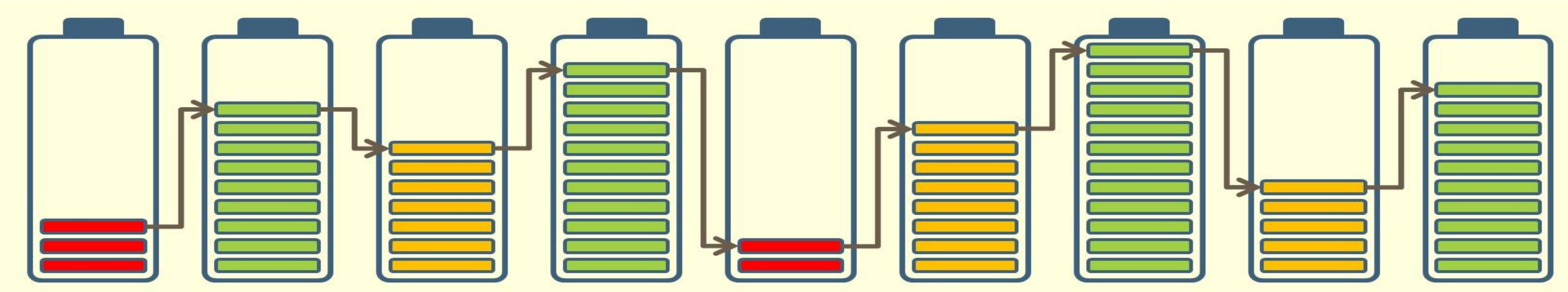
# Control accuracy



Libergreen

# 3. Cell selection



Type of vehicle; Degree of electrification; Electric driving range; Vehicle weight;
Roll and drag resistance; Usage profile; Auxillary loads; etc.

Energy and Power Requirements

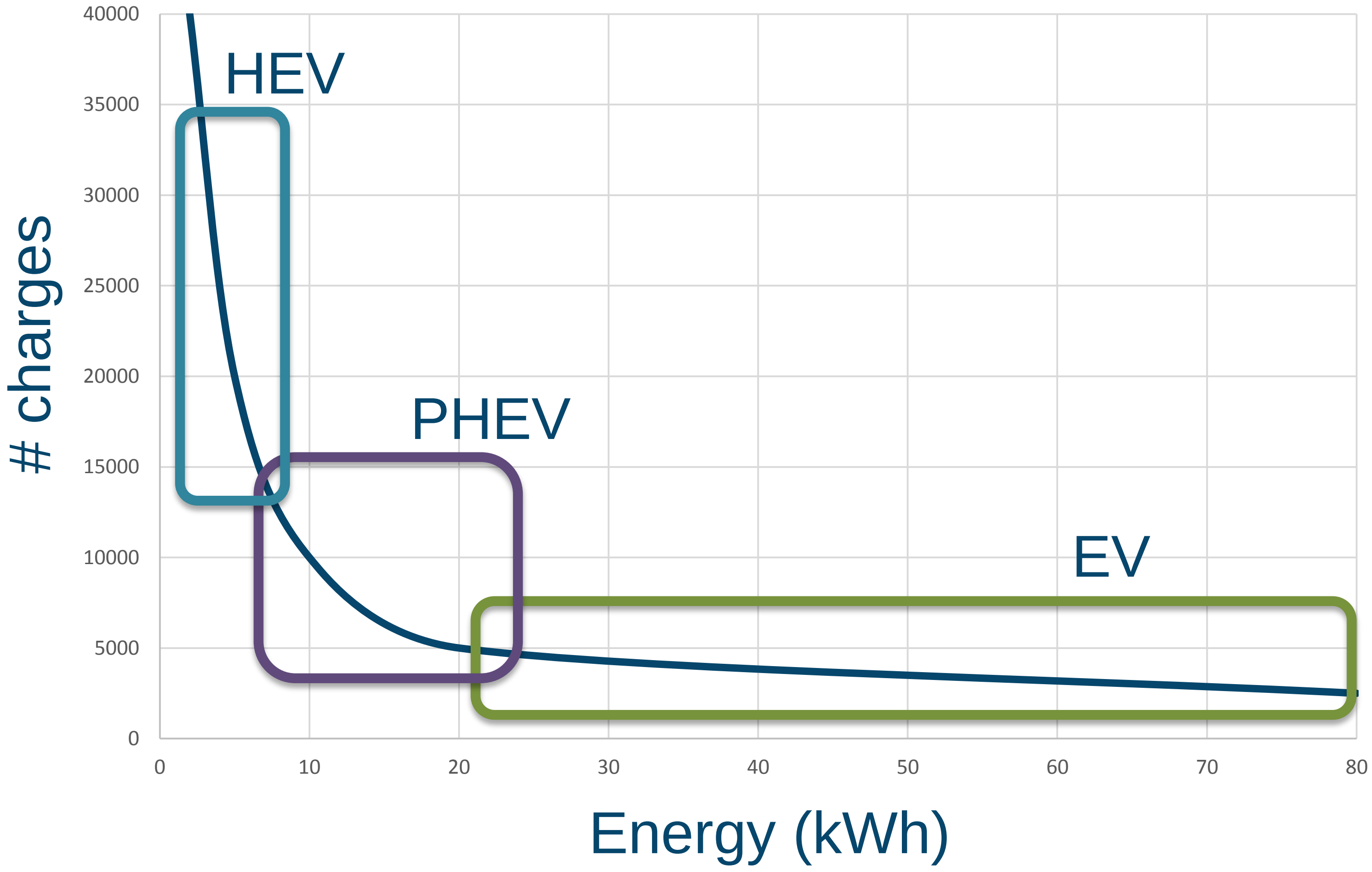
+

Packageing and climate constraints

Battery weight & volume; Voltage and SOC range;
Operational conditions; Durability; Cost; Service needs; etc.

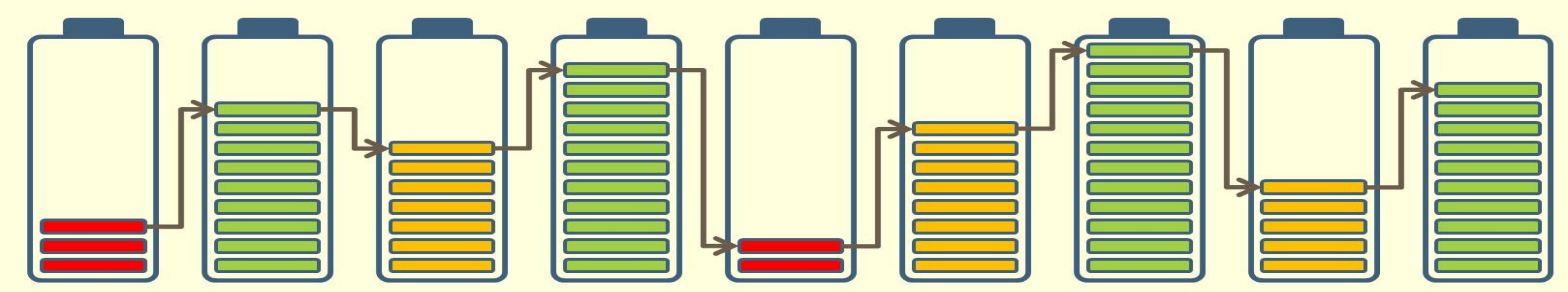
 

# Charging



40000
35000
30000
25000
20000
15000
10000
5000
0
0
10
20
30
40
50
60
70
80
# charges
Energy (kWh)
HEV
PHEV
EV

# 4. Material requirements



# Energy

500 Wh/kg → 2 km

250 Wh/kg → 1 km

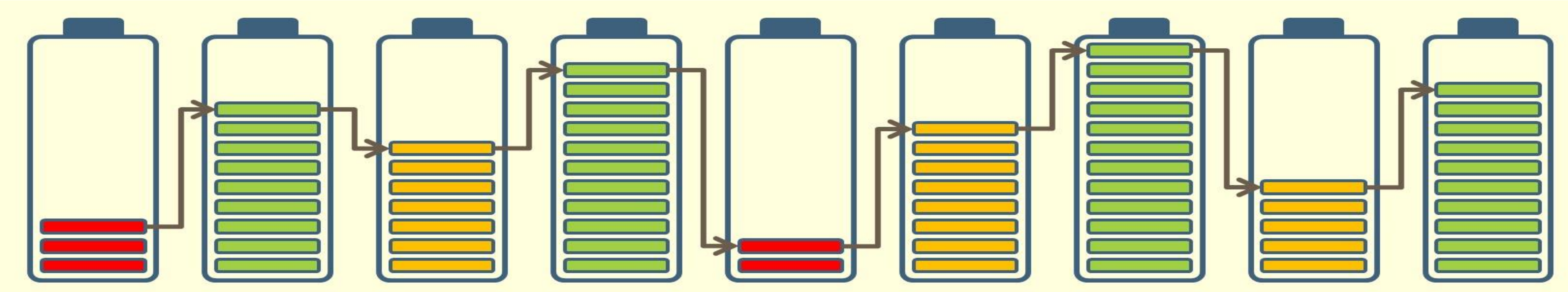
125 Wh/kg → 0.5 km

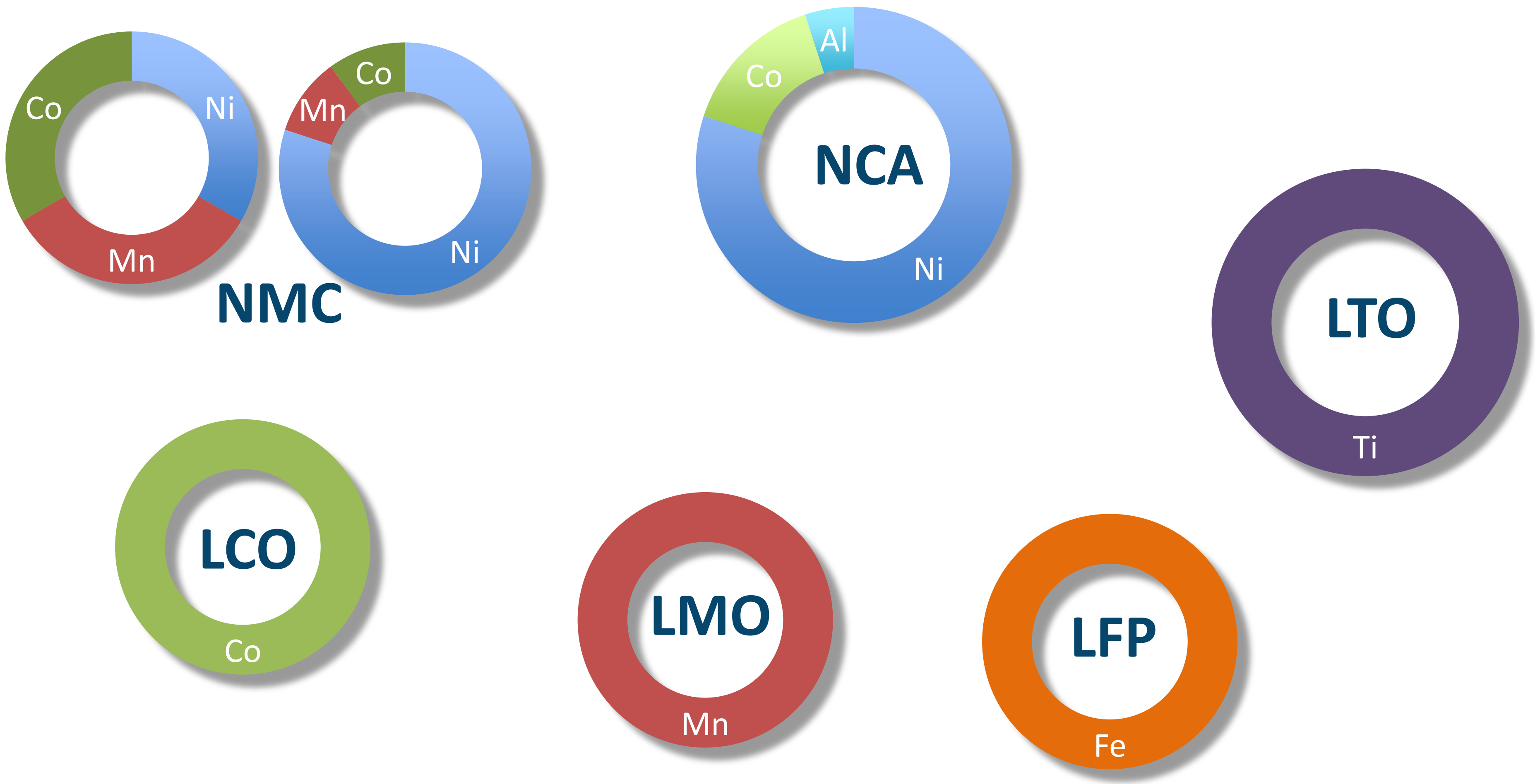
 

# Material properties needed

- High capacity (mAh/g) → high energy density
- Low impedance → good power
- High ion conductivity → fast-charging and accelerations
- Stable materials within wide potential and temperature ranges
- Number of charges → long durability
- Sustainable materials and production proceses → low cost?
- Stable in air → handling and cell production
- Low toxicity → handling
- Low-cost materials → low-cost cells?
- …

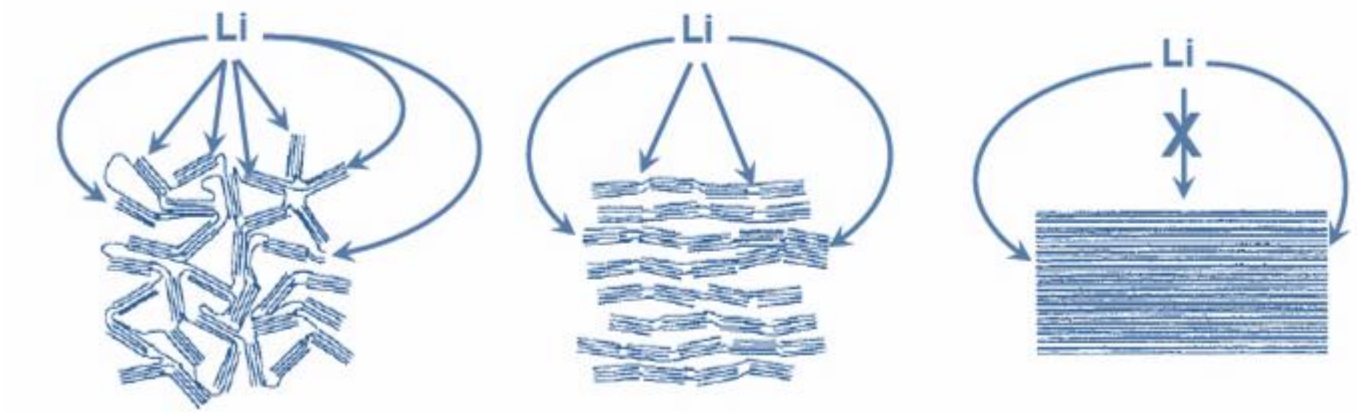
# 5. Li-ion materials




Co
Ni
Mn
Co
Mn
Ni
NMC
Al
Co
NCA
Ni
LTO
Ti
LCO
Co
LMO
Mn
LFP
Fe

# Anodes: carbons most commonly used

Hard Carbon　　Soft Carbon　　Graphite



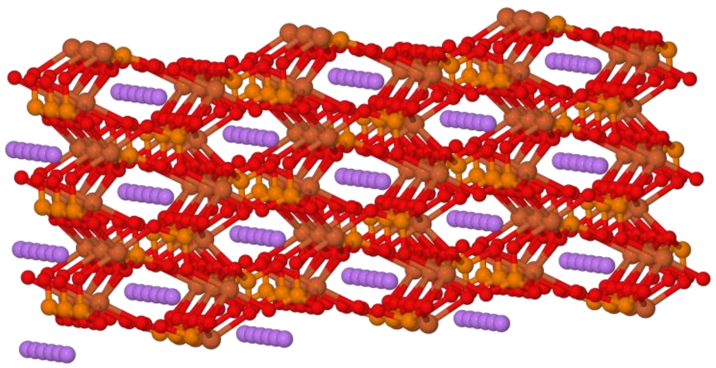

High power ←- - - - - - - - - - - - - - - - -→ Low power

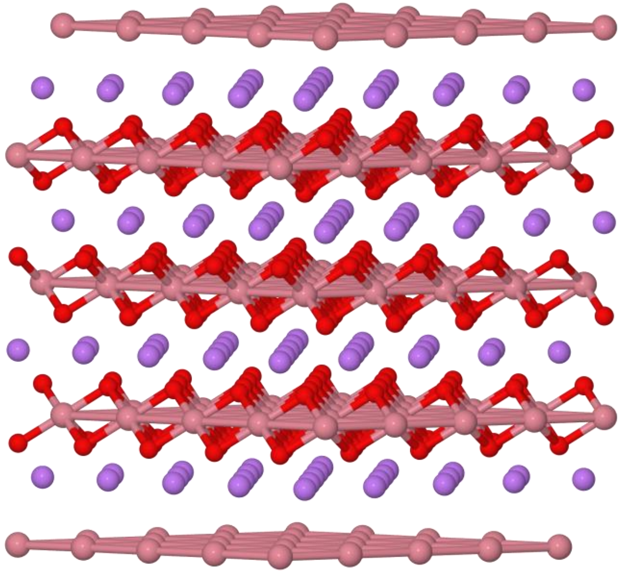
# Anodes: comparisons

| Material | Energy | Power | Safety | Cycling stability | Cost (per Ah) |
|---|---|---|---|---|---|
| Graphite | | | | | |
| Hard carbon | | | | | |
| LTO | | | | | |
| Si | | | | | |
| Li-metal | | | | | |

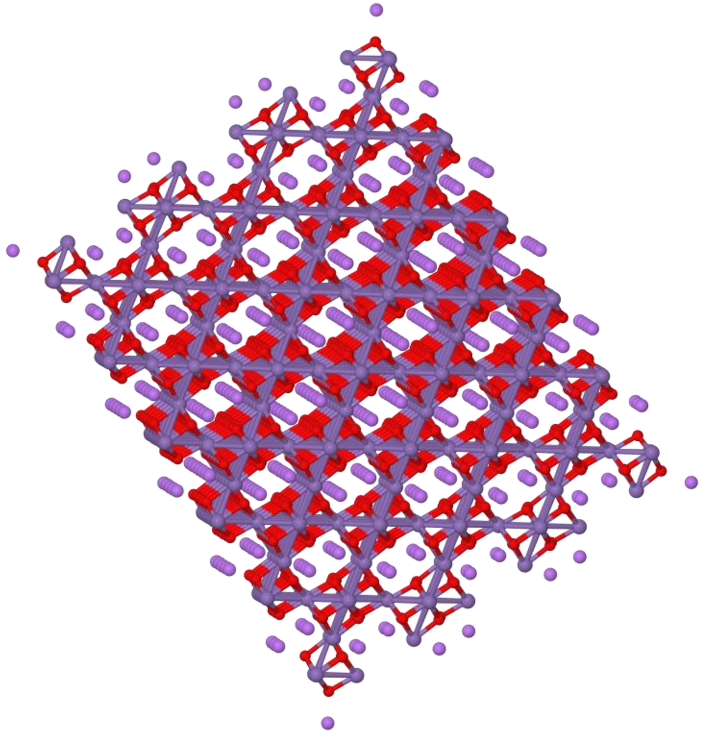
# Cathodes: transition metal oxides/phosphates



$LiFePO_4$



$LiCoO_2$, $Li(NiCoAl)O_2$, $Li(NiMnCo)O_2$

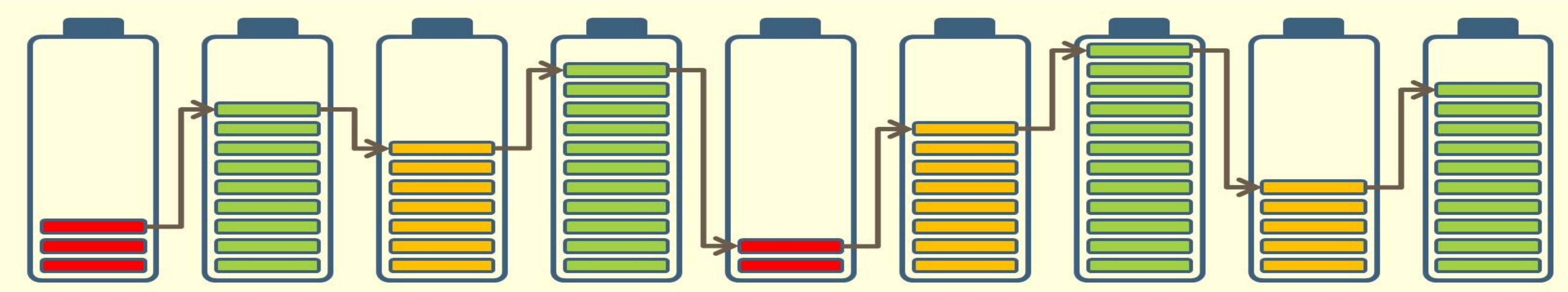


$LiMn_2O_4$

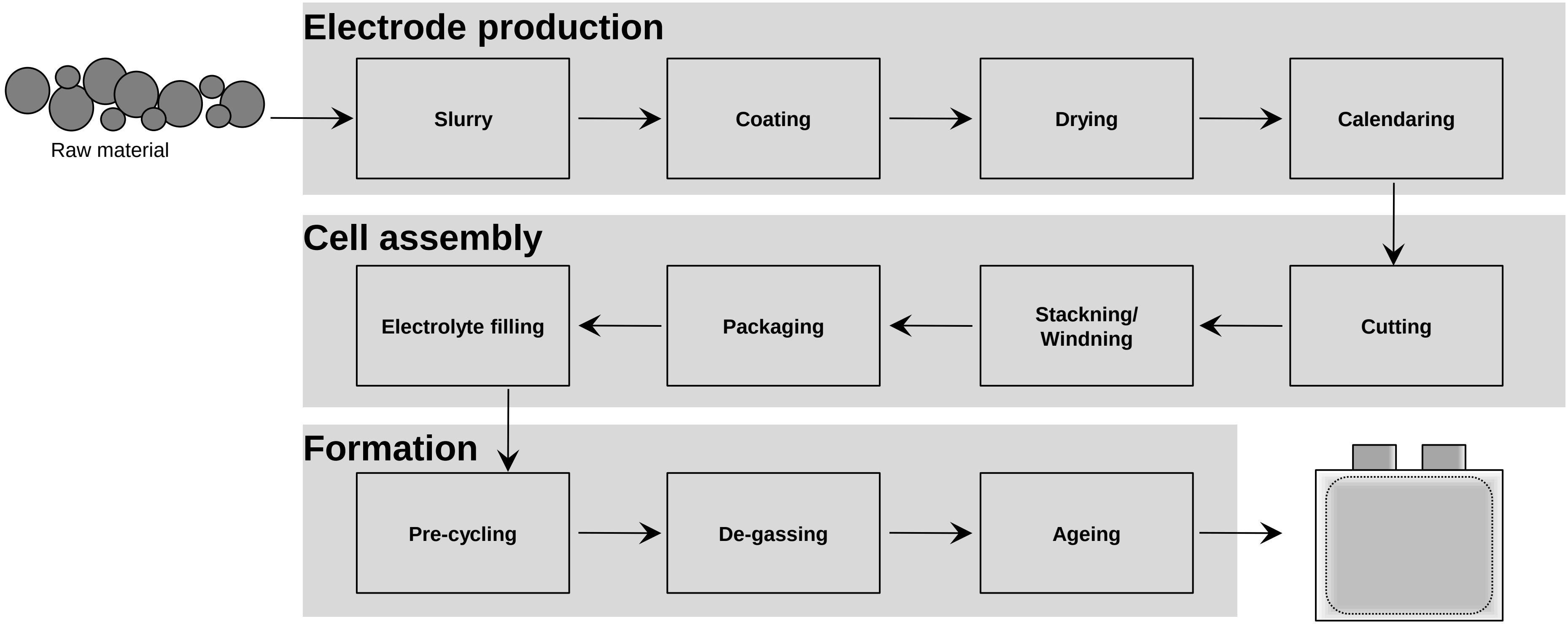
# Cathods: comparisons

| Material | Energy | Power | Safety | Cycling stability | Cost (per Ah) |
|---|---|---|---|---|---|
| LCO | | | | | |
| NCA | | | | | |
| NMC | | | | | |
| LMO | | | | | |
| LFP | | | | | |
| LMO-NMC | | | | | |

# 6. From material to cell

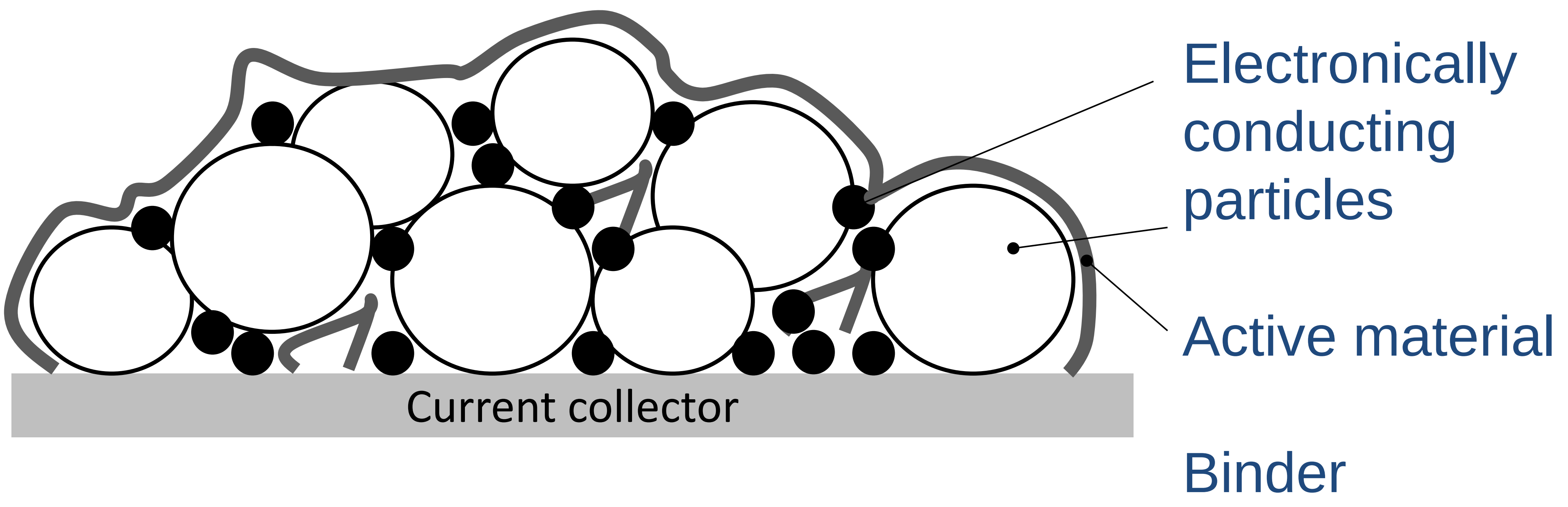


# Cell production



# Electrode design

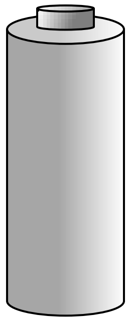


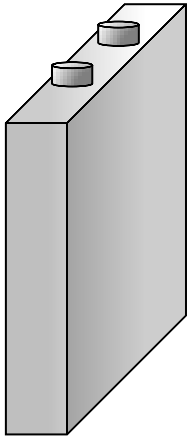

Thickness: ~50-250 μm
Loading: ~5-50 $mg/cm^2$
Porosity: ~40%

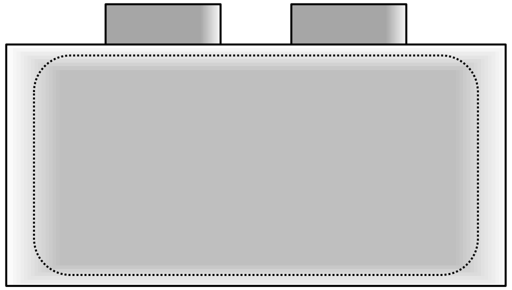


Libergreen

# Cell format







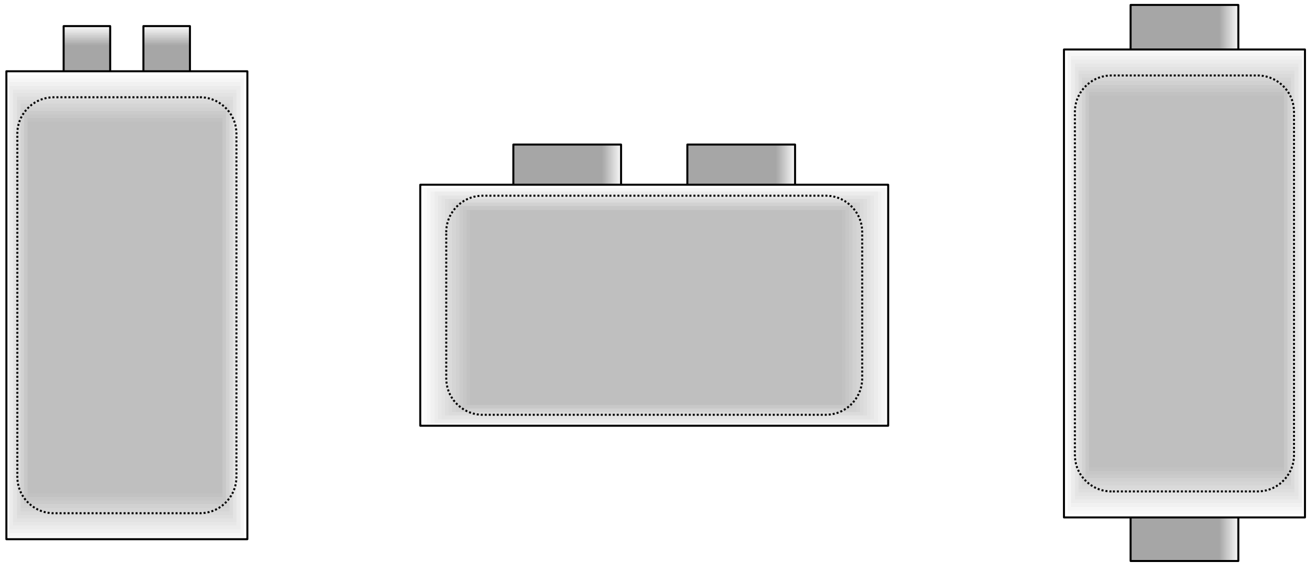
Cylindrical

Ex 18650

Prismatic

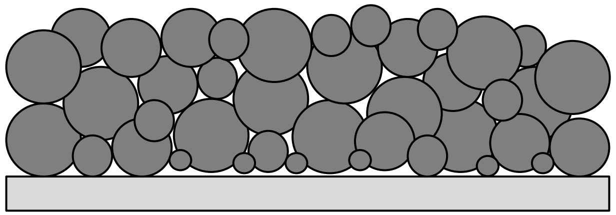
Pouch

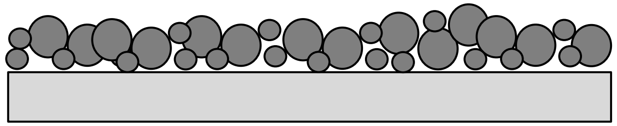
# Cell format



# Energy or power optimised cells





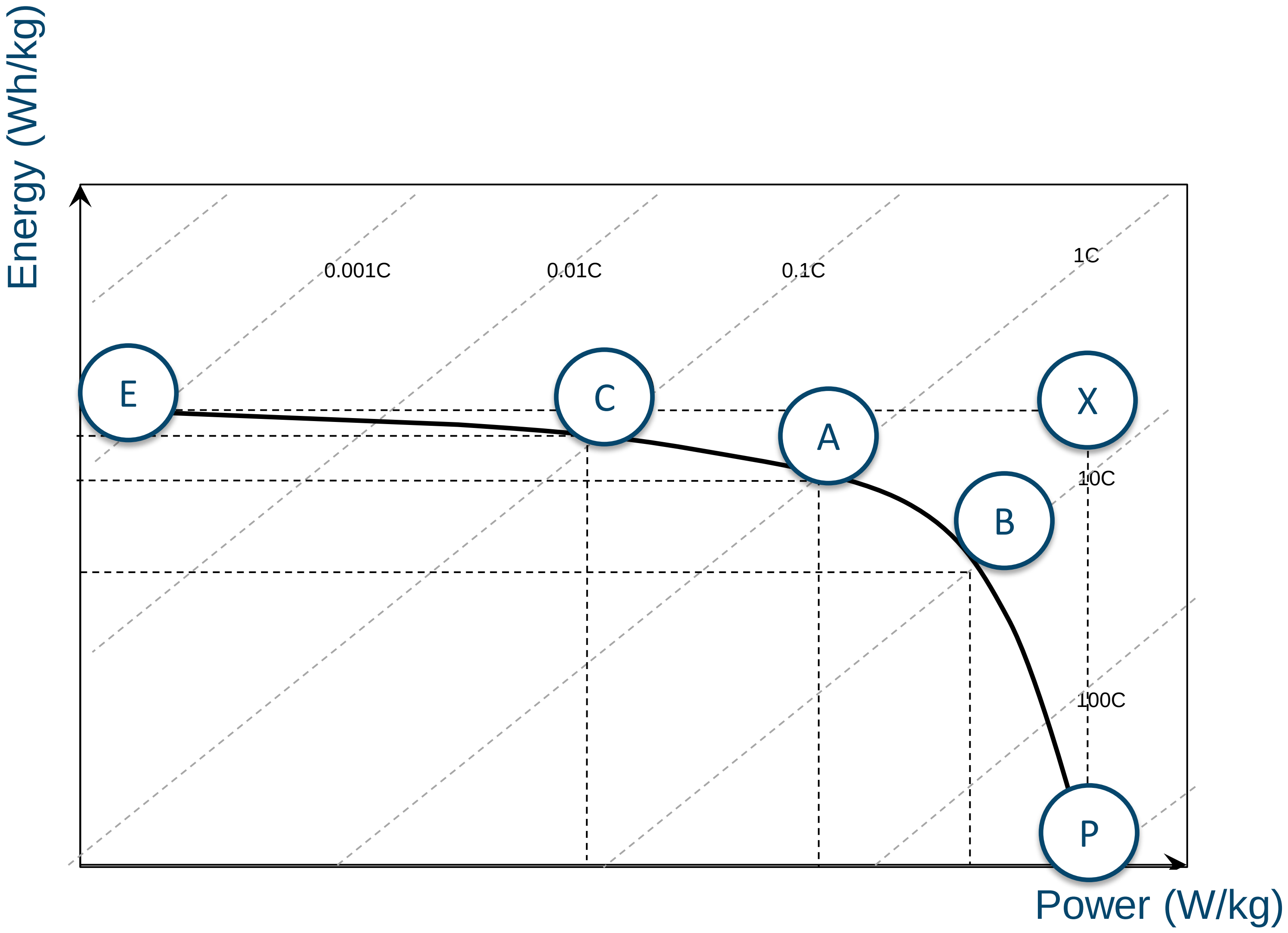
- Higher loading per cm$^2$
- Low currents to enable mass transfer, solid state diffusion, …

- Smaller particles
- Thicker current collectors to enable high currents (temperature issues if too thin)



Libergreen

# Ragone plot








Libergreen

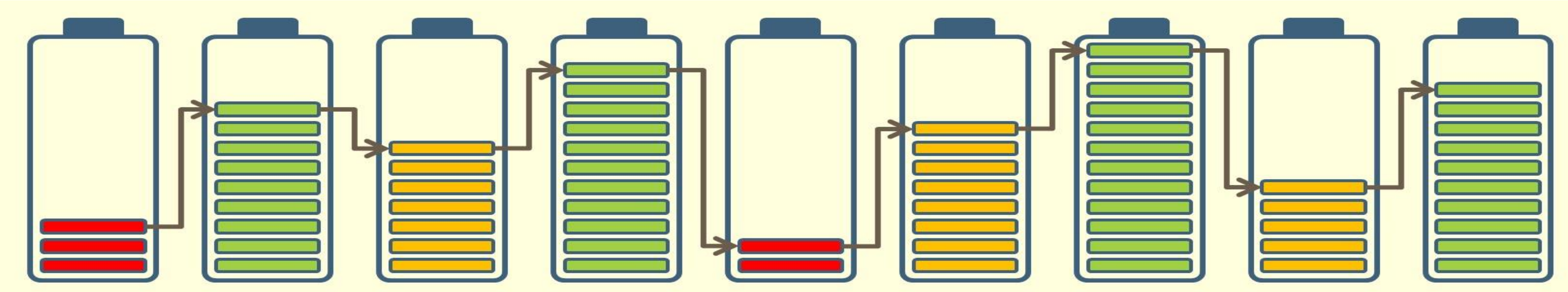
# Energy & Power

- How to increase Energy?
  - Increase the cell voltage
  - Use a cathode and anode materials with higher reversible capacity
  - Use a concept enabling higher cell voltages
  - Use concepts involving multivalent charge carriers

- How to increase Power?
  - Active materials with high Li ion diffusion
  - Electrode design – thickness, porosity, conductivity, particle size
  - Minimise resistance in materials, electrodes, cell and battery
  - Select active materials enabling fast ion diffusion

→ All related to material properties

# 7. From research to production



2018 — 2023 — 2026 — 2030

Cell development

Adv. Engineering

Product development

Production-ready cells for given application

Production-ready cells

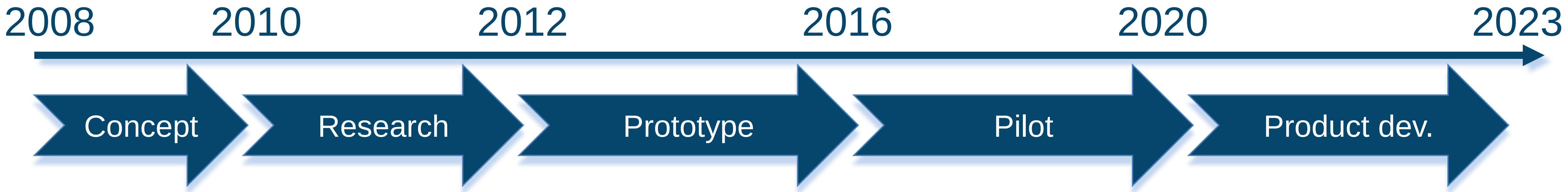
2008 | 2010 | 2012 | 2016 | 2020 | 2023

| Concept | Research | Prototype | Pilot | Product dev. |
|---|---|---|---|---|
| Re-producable results. | Fundamental understanding of performance. | Prototypes in lab. Up-scaling of materials. | Defined cell for given application. Durability understanding. | Cell designed for production. Up-scaling of production processes. |
| 1-3 years | 1-3 years | 3-5 years | 3-5 years | 2-4 years |

- Li/Na-air
- Ca, Al

- Solid-state
- Mg

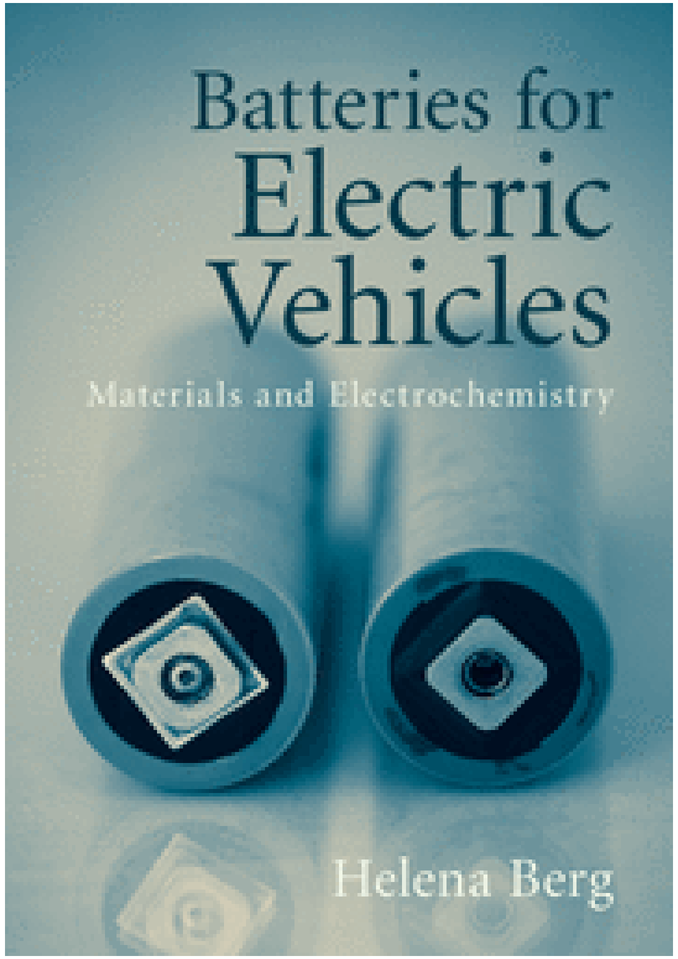
- Improved Li-ion
- Na-ion
- Li-S



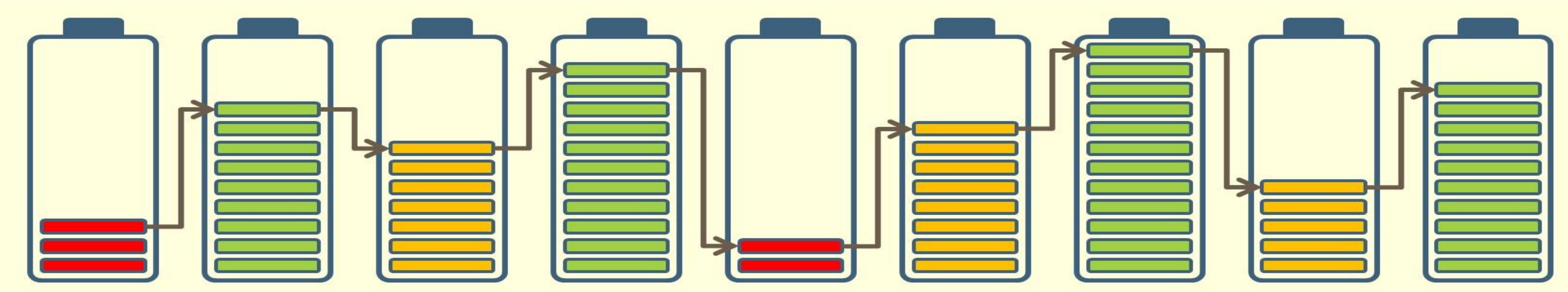

- Production processes
  - Energy usage, etc.
- Minimize inactive materials
- Optimize specific properties
  - <u>Optimize material properties</u>
- Application-adapted cells

# Further reading…



# Thank you…






Libergreen